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An Intelligent Machine Learning Approach for Prediction of Road Accident Severity

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***Abstract*—Road accidents are a primary global concern for public safety, with India having a very high death toll. This study presents an intelligent machine learning approach to predict the severity of road accidents, contributing to the development of more effective and preventive measures. We utilized a comprehensive dataset of road accidents in India, employing advanced machine learning algorithms, including Histogram-based Gradient Boosting (HGB), Random Forest (RF), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN). Our methodology involved extensive data pre-processing, feature engineering, and model optimization. The Random Forest algorithm demonstrated superior performance, achieving an accuracy of 94.7% and an F1-score of 0.947. This research provides a robust predictive model and information on the key factors influencing the accident's severity. The novelty of our approach lies in the comprehensive feature engineering tailored to the Indian context and the application of ensemble learning techniques to capture complex, nonlinear relationships in accident data. The findings of this study have significant implications for policymakers, urban planners, and law enforcement agencies in their efforts to improve road safety.**

***Index Terms*—Road accident severity, Machine learning, Random Forest, Predictive modeling, Road safety, India**

I. Introduction

Road traffic accidents result in a considerable loss of life, serious injuries, and financial hardship, making them a significant worldwide public health hazard. The World Health Organization estimates that traffic accidents claim the lives of 1.3 million people annually [1]. The issue is especially concerning in India. According to the Ministry of Road Transport and Highways, road accidents claimed the lives of around 168,491 people in India in 2022 alone [2], or 1.41 per cent of all accident-related fatalities globally.

Road accidents may range widely in severity, from minor injuries to deaths. To create efficient preventative measures and enhance road safety, it is essential to comprehend the components contributing to accident severity. Although accident data has historically been analyzed using traditional statistical approaches, these methods often fail to capture the intricate, non-linear correlations between many variables and accident severity.

In recent years, machine learning techniques have become more effective tools for studying vast and complicated information. They can reveal links and patterns that previous approaches may have overlooked. Due to their ability to handle several variables simultaneously, consider interactions and non-linear correlations, and adjust to new information, these methods are especially well-suited for the dynamic nature of road accident investigation.

Our study aims to use machine learning to create a clever method for estimating the severity of traffic accidents in India. This method may help minimize the frequency and consequences of severe traffic accidents by providing precise estimates of the potential severity of incidents. It can also help with resource allocation, targeted interventions, and policy choices.

The main contributions of this study are:

1) Creation of a thorough machine learning framework that uses sophisticated algorithms and intensive feature engineering to forecast the severity of traffic accidents.

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2) To find the best solution, many machine learning techniques are compared, including K-Nearest Neighbors, Random Forest, Support Vector Machines, and Gradient Boosting based on histograms.
3) Insights into the main variables affecting the severity of traffic accidents in India may help with evidence-based legislation and focused initiatives.
4) A robust prediction model that allows authorities to evaluate the probable severity of accidents and take preventative action to reduce risks.

The remainder of this paper is structured into several sections: Section II provides an overview of relevant research on predicting the severity of traffic accidents. Section III describes the dataset and technique used in this work, including feature engineering, data preprocessing, and model construction. Section IV shows the outcomes of our tests and explains the results. Section V concludes the paper with further study recommendations.

II. Related Work

In recent years, there has been a noticeable increase in interest in using machine learning methods to forecast the severity of road accidents. Numerous investigations have examined diverse algorithms and methodologies to enhance forecasts' precision and dependability.

Zhu et al. [3] employed a random forest algorithm to predict the severity of traffic accidents on urban expressways in Shanghai. Their study highlighted the importance of vehicle type, driver age, and weather conditions in determining accident severity. Similarly, Kalra et al. [4] used a combination of decision trees and random forests to analyze road accident data in India, achieving an accuracy 87% in predicting accident severity.

Support Vector Machines (SVM) have also been widely used in this domain. Alkheder et al. [5] applied SVM to predict injury severity in traffic accidents in Dubai, demonstrating the algorithm's effectiveness in handling non-linear relationships between variables. Their model achieved an accuracy of 74.6% in classifying accident severity.

Deep learning approaches have shown promise in recent studies. Bao et al. [6] used a Deep Neural Network (DNN) to predict traffic accident duration, often correlated with severity. Their model performed better than conventional machine learning techniques, demonstrating the use of deep learning in this domain.

Ensemble methods have gained popularity due to their ability to combine multiple models and reduce overfitting. Pande et al. [7] used a stacking ensemble method, combining decision trees, random forests, and gradient-boosting machines to predict highway crash severity. Their approach showed improved performance compared to individual algorithms.

Even though these researches have made a big difference, predicting accuracy and model interpretability may still be improved. Furthermore, a customized strategy is required due to the distinct features of traffic accidents in India, including various vehicle kinds and road conditions. In order to obtain a better prediction of accuracy for road accident severity in the Indian context, our research expands on this body of work by offering a complete framework that includes sophisticated preprocessing approaches, feature engineering, and cutting-edge machine learning algorithms.

III. Methodology

A. Dataset Description

The dataset used in this study is sourced from the Road Accident Severity in India dataset [8], which includes extensive data on road accidents in India. The dataset includes 2890 categories with 23 attributes spanning a range of accident-related areas, including temporal factors, driver characteristics, vehicle information, road conditions, and environmental factors. Key features in the dataset include:

- Temporal information: Time, Day of week
- Driver characteristics: Age band, Sex, Educational level, Driving experience
- Vehicle information: Type of vehicle, Owner of vehicle, Service year of vehicle
- Road conditions: Area accident occurred, Road alignment, Types of junction, Road surface type
- Environmental factors: Light conditions, Weather conditions
- Accident details: Type of collision, Vehicle movement, Casualty class, Cause of accident

Table I presents the encoding scheme used for the features in our analysis.

TABLE I
Feature Encoding Scheme for Road Accident Severity Dataset [8]

Feature	Encoding
Time	Hour extracted (0-23)
Day_of_week	Monday(0) to Sunday(6)
Age_band_of_driver	Under 18(0), 18-30(1), 31-50(2), Above 50(3)
Sex_of_driver	Label Encoded
Educational_level	Elementary(0), Junior high(1), High school(2), College(3)
Driving_experience	1-2yr(1), 2-5yr(2), 5-10yr(3), Above 10yr(4)
Type_of_vehicle	Automobile(0), Lorry(1), Public(2), Motorcycle(3)
Owner_of_vehicle	Owner(0), Others(-1)
Service_year_of_vehicle	1-2yr(1), 2-5yrs(2), 5-10yrs(3), Above 10yr(4)
Defect_of_vehicle	No defect(0), Defect(1)
Area_accident_occured	Residential(0), Office(1), Industrial(2)
Road_allignment	Tangent flat(0), Escarpments(1)
Types_of_Junction	No junction(0), Y Shape(1), T Shape(2)
Road_surface_type	Asphalt(0), Earth(1)
Road_surface_conditions	Dry(0), Wet(1)
Light_conditions	Daylight(0), Others(-1)
Weather_conditions	Normal(0), Others(-1)
Type_of_collision	Parked vehicles(0), Animals(1)
Vehicle_movement	Straight(0), U-Turn(1), Waiting(2)
Cause_of_accident	Lane change right(0), Lane change left(1)
Casualty_class	Pedestrian(0), Passenger(1), Driver(2)
Casualty_severity*	Slight(1), Serious(2), Fatal(3)
Work_of_casuality	Driver(0), Passenger(1), Pedestrian(2)
Pedestrian_movement	Crossing nearside(0), Not pedestrian(2)

*Target Variable

The target variable is 'Accident_severity', which is categorized into three levels: Slight Injury, Serious Injury, and Fatal Injury.

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B. Data Preprocessing

In order to guarantee the dependability and quality of the machine learning models, data preparation is an essential step. Our preprocessing pipeline includes the following steps:

1) Handling missing values: We removed columns with a high percentage of missing data and used appropriate imputation techniques for columns with a lower percentage of missing values.
2) Feature encoding: Categorical variables were encoded using a label or one-hot encoding, depending on their cardinality and ordinal nature.
3) Feature scaling: Numerical features were standardized using the StandardScaler to ensure they are all on the same scale, which is particularly important for algorithms like SVM and KNN.
4) Temporal feature engineering: We extracted the hour from the 'Time' column to capture potential patterns related to the time of day.
5) Handling imbalanced data: We have utilized the Synthetic Minority Over-sampling Technique (SMOTE) to resolve the class imbalance in the target variable, guaranteeing our models are not skewed towards the majority class.

C. Feature Selection and Engineering

Feature selection and engineering are vital in improving model performance and interpretability. We employed the following techniques:

1) Correlation analysis: We calculated the correlation between features and removed highly correlated features to reduce multicollinearity.
2) Feature importance: We determined which characteristics were most important in forecasting the severity of accidents using the Random Forest algorithm's feature importance scores.
3) Domain-specific feature engineering: Based on domain knowledge, we created new features such as 'Rush_hour' (binary feature indicating peak traffic hours) and 'Weekend' (binary feature for weekends).

D. Model Development and Evaluation

We implemented and compared four machine learning algorithms:

1) Histogram-based Gradient Boosting (HGB)
2) Random Forest (RF)
3) Support Vector Machines (SVM)
4) K-Nearest Neighbors (KNN)

For each algorithm, we performed hyperparameter tuning using GridSearchCV with 5-fold cross-validation to find the optimal set of parameters. These models were trained on 80% of the data and tested on the remaining 20%.

We evaluated the models using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

Additionally, we generated confusion matrices for each model to provide a detailed view of their performance across different severity classes.

E. System Architecture

Fig. 1 illustrates the architecture of our proposed system for road accident severity prediction.

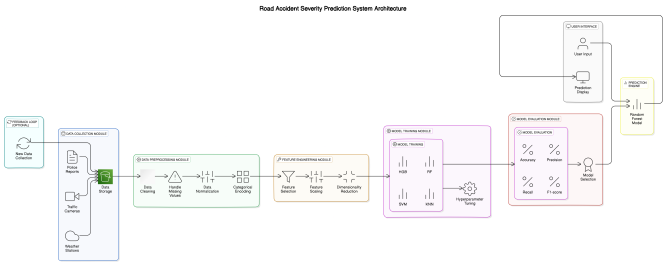


Fig. 1. Architecture of the proposed road accident severity prediction system

F. Algorithm

The following algorithm outlines the key steps in our methodology:

Algorithm 1 Road Accident Severity Prediction

Require: Road accident dataset D
Ensure: Trained model M, Evaluation metrics E

```
D_preprocessed ← Preprocess(D)
D_train, D_test ← SplitData(D_preprocessed, ratio=0.8)
D_train_balanced ← ApplySMOTE(D_train)
for each algorithm A in [HGB, RF, SVM, KNN] do
    params ← PerformGridSearch(A, D_train_balanced)
    M_A ← TrainModel(A, D_train_balanced, params)
    predictions ← PredictModel(M_A, D_test)
    E_A ← EvaluateModel(predictions, D_test)
end for
M_best ← SelectBestModel(E)
return M_best, E =0
```

IV. Results and Discussion

A. Model Performance

Table I presents the performance metrics of the four algorithms tested in this study.

TABLE II
Performance Metrics of Different Algorithms

Algorithm	Accuracy	Precision	Recall	F1-score
HGB	0.938	0.943	0.938	0.939
RF	0.947	0.954	0.947	0.947
SVM	0.937	0.942	0.937	0.937
KNN	0.868	0.894	0.872	0.863

The Random Forest algorithm demonstrated the best overall performance, achieving the highest scores across all metrics. It outperformed the other algorithms with an accuracy of 94.7%, precision of 95.4%, recall of 94.7%, and an F1-score of 0.947.

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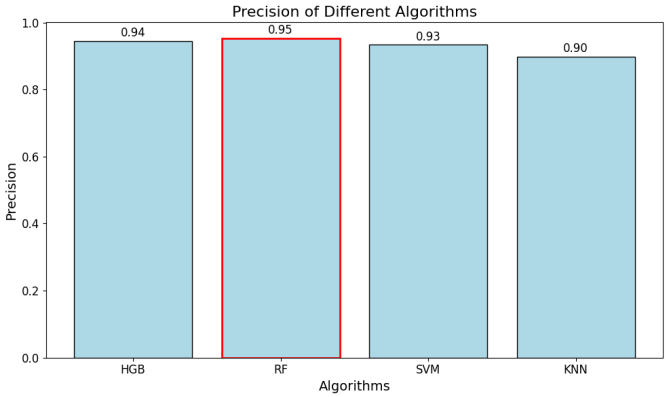


Fig. 2. Comparison of precision of different algorithms

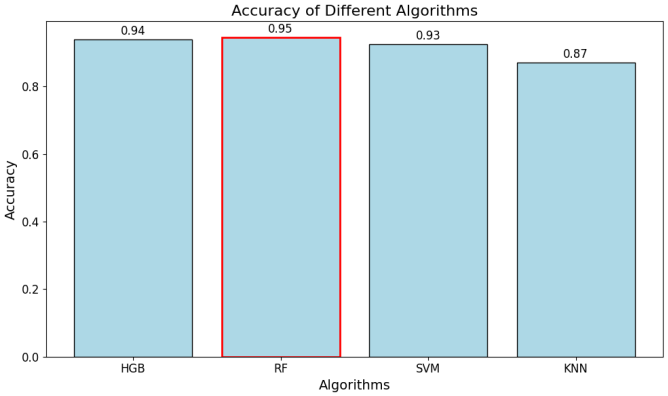


Fig. 3. Comparison of accuracy of different algorithms

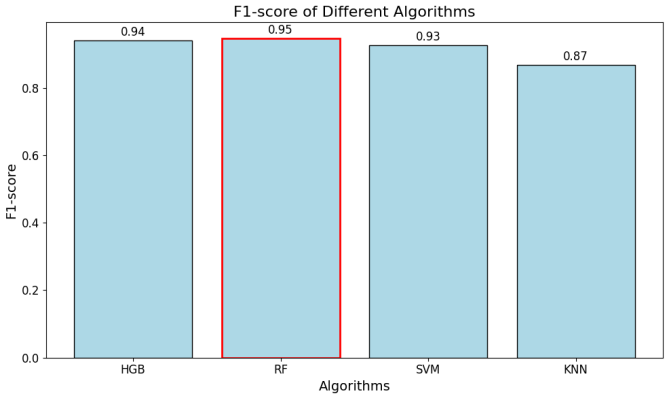


Fig. 4. Comparison of F1-score of different algorithms

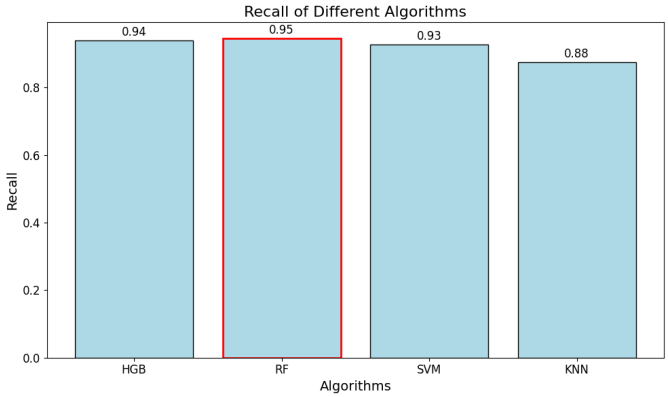


Fig. 5. Comparison of recall of different algorithms

B. Feature Importance

Analysis of feature importance from the Random Forest model revealed the top factors influencing accident severity:

1) Types of Junction
2) Day of Week
3) Age Band of Driver
4) Light Conditions
5) Weather Conditions

This information can be valuable for policymakers and urban planners when designing targeted interventions to reduce severe accidents.

C. Discussion

The Random Forest algorithm performs better than others because it can handle non-linear interactions and is resistant to overfitting. The ensemble nature of Random Forests, combining multiple decision trees, allows it to capture complex patterns in the data that single-model approaches like SVM or KNN might miss.

Our model's high accuracy of 94.7% marks a notable advancement above earlier research in this subject. For instance, Kalra et al. [4] reported an accuracy of 87% using a combination of decision trees and random forests on Indian road accident data. Our improved performance can be attributed to:

1) Comprehensive feature engineering, including the creation of domain-specific features.
2) Effective handling of class imbalance using SMOTE.
3) Extensive hyperparameter tuning using GridSearchCV.

Identifying key factors influencing accident severity aligns with findings from previous studies. For example, Zhu et al. [3] also found driver age and weather conditions essential factors in accident severity prediction.

Figures 6-9 present the confusion matrices for each algorithm tested. These matrices provide a detailed view of each model's performance across different severity classes, highlighting areas where each algorithm excels or struggles.

V. Conclusion and Future Work

This study presents a comprehensive machine-learning approach for predicting road accident severity in India. We

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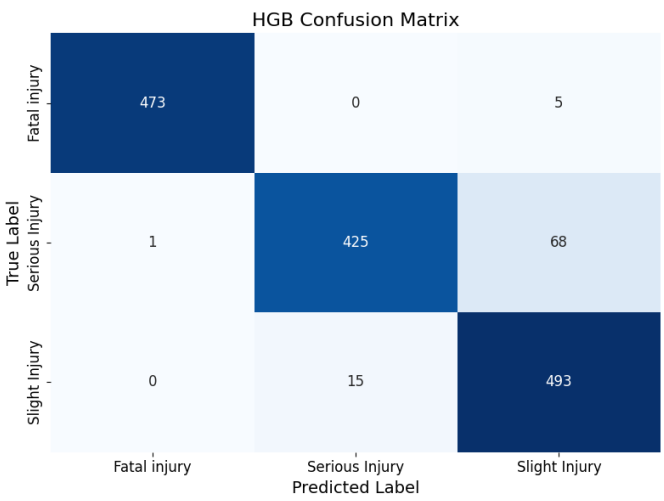


Fig. 6. Confusion matrix of HGB

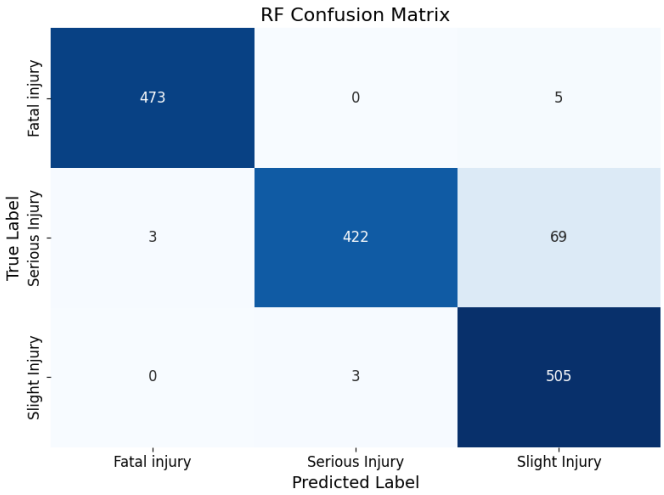


Fig. 7. Confusion matrix of RF

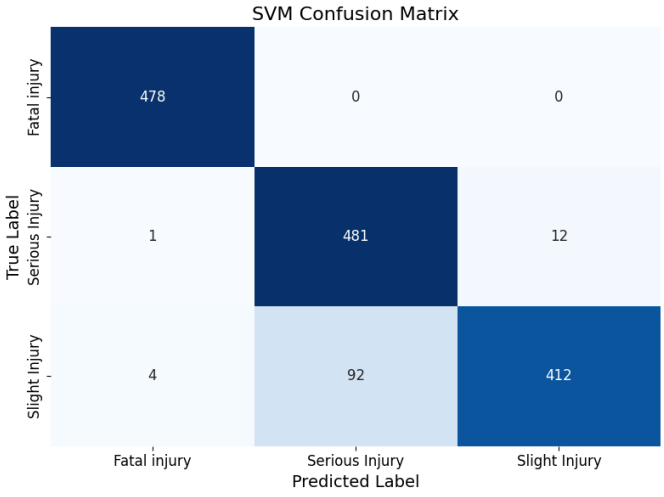


Fig. 8. Confusion matrix of SVM

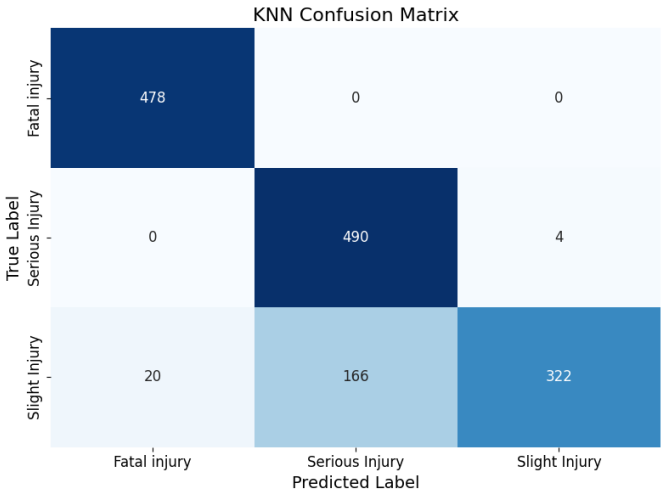


Fig. 9. Confusion matrix of KNN

have developed a model that accurately calculates accident severity by leveraging advanced preprocessing techniques, feature engineering, and state-of-the-art algorithms.

The Random Forest algorithm emerged as the most effective method, demonstrating superior performance across all evaluation metrics. This model provides accurate predictions and insights into the key factors influencing accident severity, such as junction types, day of the week, driver age, and environmental conditions.

These findings have significant implications for road safety policy and urban planning. By identifying the most influential factors in accident severity, policymakers and urban planners can develop targeted interventions to reduce the incidence of severe accidents. This could include redesigning high-risk junctions, implementing age-specific driver education programs, improving road lighting infrastructure, and developing weather-based traffic management systems.

While our model shows promising results, there are several avenues for future research and improvement:

1) Real-time prediction: Developing a system that integrates live traffic and weather data for real-time accident severity prediction.
2) Deep learning approaches: As the dataset grows, exploring deep learning methods such as neural networks might capture even more complex patterns in the data.
3) Transfer learning: Investigating the potential of applying the model to different geographical regions or countries.
4) Interpretability: Employing techniques like SHAP (Shapley Additive exPlanations) values could provide more detailed, instance-level explanations of predictions.
5) Multimodal data: Incorporating data from additional sources, such as vehicle telematics or road sensor data, could provide a more comprehensive view of accident risk factors.

In conclusion, this study demonstrates the potential of machine learning in addressing the critical issue of road safety. By

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accurately predicting accident severity and identifying key risk factors. Our approach can contribute to developing data-driven strategies to reduce severe accidents, ultimately saving lives and mitigating the socio-economic impact of road accidents in India and beyond.

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